AI Partner Churn Prediction & Early Warning System

The Challenge

How might we build an AI system that predicts which partners are at risk of leaving, identifies why they're disengaging, and alerts relationship managers before it's too late?

The Problem

Partner churn is invisible until it's irreversible. By the time you notice a partner has stopped sending traffic or is promoting competitors, months of revenue are gone and the relationship is already dead.

> *"Our top partner in the region went from high performer to barely active over a few months. We didn't notice until it was too late."*
>
> *"Partner stopped logging into the dashboard weeks ago. No one flagged it. Now they're promoting a competitor."*
>
> *"We have thousands of active partners. I manage hundreds personally. How do I know which ones need attention this week?"*
>
> *"Partner submitted multiple support tickets about commission discrepancies. No one connected the dots. A few months later, they churned."*

The core issue: relationship managers are reactive, not proactive. They only notice problems when partners explicitly complain or when performance has already collapsed. Silent disengagement goes undetected.

Why This Matters Now

Affiliate and partner networks drive a significant portion of customer acquisition for most platforms. Partner churn is expensive:

- Recruiting costs significantly more than retention
- Revenue loss is delayed: Partners slowly reduce activity before leaving completely
- High-value churn hurts most: A small percentage of top partners often generate the majority of referrals
- Competitive poaching: Rivals actively recruit your best performers
- Scale problem: Can't personally monitor thousands of relationships

Traditional CRM doesn't help because it tracks what happened, not what's about to happen. Managers need predictive intelligence.

The Opportunity

Build an AI-powered partner churn prediction system that:

Early Warning Detection:

- Activity pattern analysis: Track partner engagement trends over time

- Login frequency declining
- Referral volume dropping (week-over-week, month-over-month)
- Traffic quality decreasing
- Commission earnings falling
- Dashboard usage patterns are changing
- Engagement degradation scoring: Quantify how partner behaviour is shifting compared to their historical baseline

Multi-Signal Risk Scoring:

- Behavioural signals: Activity decline, engagement drop-off, performance degradation
- Communication signals: Support ticket volume/sentiment, response time to outreach, feedback tone
- Relationship signals: Event attendance, training participation, program adoption
- Financial signals: Commission trends, payout issues, payment disputes
- Competitive signals: Mentions of competitors, pricing complaints, feature requests

Churn Probability Prediction:

- Risk scoring: Assign churn probability to each partner (High / Medium / Low risk tiers)
- Explainable predictions: Clear reasoning for risk scores
- Example: "High risk because: Login frequency declining significantly, referral volume down, unresolved support tickets, competitor mentioned in recent feedback"

Intelligent Alerting & Prioritisation:

- Real-time alerts: Notify partner managers when risk scores cross thresholds
- Priority ranking: Focus manager attention on highest-value, highest-risk partners
- Segment analysis: Identify when multiple partners in a region or segment show similar churn signals - possible systemic issue

The system should detect disengagement well before churn occurs, with enough lead time for effective intervention.

Constraints

Constraint	Rationale
Must demo live	Show actual churn prediction on sample partner data.
AI must add value	This is an AI hackathon. GenAI must be core to your solution.
Explainable predictions	Managers need to understand WHY someone is at risk.
Minimise false positives	Alert fatigue damages manager effectiveness.
Actionable insights	Risk scores alone aren't enough - need context for intervention.

Evaluation Criteria

Criterion	Weight	What We're Looking For
Insight	30%	Does it correctly identify partners who will churn?
Usefulness	25%	Can it predict well in advance with useful lead time for intervention?
Craft	20%	Does it find patterns humans would miss? Is the UX intuitive?
Ambition	15%	Did you take risks? Try something unexpected?
Demo	10%	Can you tell the story of what you built and why it matters?

Questions Worth Considering

- What behavioural signals are most predictive of churn?
- How do you distinguish temporary performance dips from genuine disengagement?
- What's the right prediction window for useful intervention?
- Should alerts be partner-specific or identify cohort-level trends?
- How do you balance sensitivity (catch all churn) vs. specificity (avoid false alarms)?
- Can the system learn which interventions successfully prevent churn?

What Would Blow Our Minds

- Accurate early prediction: High churn probability identified with clear reasoning based on multiple declining signals
- Cohort risk detection: "Warning: Multiple partners in this region showing similar churn signals - investigate systemic issue"
- Intervention timing: Identifying the optimal intervention window before churn probability becomes too high
- Pattern discovery: "New churn pattern detected: partners with unresolved technical issues for extended periods have significantly higher churn rates"
- Learning from outcomes: System that improves predictions based on which partners actually churned vs. were successfully retained
- Proactive recommendations: Not just alerting to risk, but suggesting specific retention actions based on the churn drivers identified